

Han--Li--Sun	β -	$\beta > 0$	weighted Gaussian density
2			
1. Han--Li--Sun	ε -		
		$\Psi_p(X_0,T;T-r^2) < 1 + \varepsilon_{\text{HLS}},$	(1.1)
	$\varepsilon_{\text{HLS}} > 0$	$\min\{\varepsilon_{\text{HLS}}, 1/2\}$	$0 < \varepsilon_{\text{HLS}} < 1$ (1 +
$\varepsilon_{\text{HLS}}, 2)$	$\Psi_p < 2$	(1.1)	
2. “ ”		2	1
	$(1, 2)$	1	
3. $\beta > 0$		White	Han--Li--Sun (1.1)

“weighted Gaussian density < 2”

(1.1) $1 + \varepsilon_{\text{HLS}}$

1.

Han--Li--Sun

$$\partial_t F = \frac{\cos^2 \alpha \, H - \beta \sin^2 \alpha \, V}{\cos^2 \alpha + \beta \sin^2 \alpha}, \qquad \beta \geq 0, \tag{1.2}$$

$$\alpha \quad \text{K\"ahler} \qquad \beta = 0 \qquad (1.2) \qquad \partial_t F = H \qquad \cos \alpha \geq \delta > 0 \qquad p \geq p_0(\beta, \delta)$$

$$\Psi_p(X_0,t_0;t) = \int_{\Sigma_t} \phi(F) \, \rho_{X_0,t_0}(F,t) \frac{1}{\cos^p \alpha} \, d\mu_t, \tag{1.3}$$

$$\phi \quad X_0 \quad 1 \quad \rho$$

“weighted Gaussian density < 2”

- $r > 0 \quad \Psi_p(X_0,T;T-r^2) < 2$
-

$$\Theta_p^w(X_0,T):=\lim_{r\downarrow 0}\Psi_p(X_0,T;T-r^2)<2. \tag{1.4}$$

$$\text{Han--Li--Sun} \tag{1.1}$$

2

$$2. \qquad 2$$

$$2 \qquad 0<\cos\alpha\leq 1$$

$$2.1 \quad D_p<2$$

$$\mathcal{C}=\sum_{j=1}^Nm_j|P_j|, \qquad m_j\in\mathbb{N}, \tag{2.1}$$

$$\text{K\"ahler} \qquad c_j=\cos\alpha_{P_j}\in(0,1]$$

$$D_p(\mathcal{C})=\sum_{j=1}^Nm_jc_j^{-p}. \tag{2.2}$$

$$D_p(\mathcal{C})<2 \qquad \sum_jm_j=1 \qquad c \tag{2.3}$$

$$c^{-p}<2 \quad \Longleftrightarrow \quad c>2^{-1/p}.$$

$$c>2^{-1/p} \qquad D_p<2 \qquad c<1 \qquad D_p=c^{-p}>1$$

$$0<c_j\leq 1 \quad p>0 \quad c_j^{-p}\geq 1$$

$$\sum_{j=1}^Nm_j\leq \sum_{j=1}^Nm_jc_j^{-p}=D_p(\mathcal{C})<2. \tag{2.4}$$

$$2 \qquad 1 \tag{2.1} \qquad 1 \qquad (2.3) \quad c^{-p}<2$$

$$D_p=c^{-p}$$

$$2.2 \qquad \text{HLS}$$

$$D_p<2 \qquad (2.2) \qquad D_p=1$$

$$[1,2) \qquad c^{-p} \qquad 0<\varepsilon<1$$

$$1+\varepsilon<c^{-p}<2 \tag{2.5}$$

$$c\in(2^{-1/p},(1+\varepsilon)^{-1/p}) \qquad D_p<2 \quad D_p>1+\varepsilon \qquad c=1$$

$$D_p<1+\varepsilon_{\text{HLS}}$$

$$c \in (0,1] \qquad \mathbb{C}^2 = \mathbb{R}^4 \qquad e_1, e_2, e_3, e_4 \qquad Je_1 = e_2 \quad Je_3 = e_4$$

$$P_c = \operatorname{span} \left\{ e_1, ce_2 + \sqrt{1-c^2} \, e_3 \right\} . \tag{2.6}$$

Kähler

$$\omega\Big(e_1, ce_2 + \sqrt{1-c^2} \, e_3\Big) = c. \tag{2.7}$$

$$P_c \quad \text{K\"ahler} \qquad c \qquad P_c \qquad A=H=0 \qquad \alpha \qquad V=0 \qquad (1.2)$$

$$“ \qquad 2 \qquad \qquad \text{HLS} \qquad \qquad ”$$

3.

$$\cos \alpha = *F_t^* \omega \geq \delta > 0, \qquad |A_t|^2 \leq \frac{C}{T-t} \tag{3.1}$$

$$\cos^{-p} \alpha$$

- Kähler
-
- $\Theta_p^w < 2 \qquad \Theta_p^w < 1 + \varepsilon_{\text{HLS}}$
- (1.2) “ ” White
- $|A|^2 \leq C/(T-t)$ Type-I

$$\begin{array}{llll} \text{Han--Li--Sun} & \text{Theorem 5.7} & \lambda- & \\ \text{“holomorphic curve in } \mathbb{C}^2\text{”} & & \text{“} & \text{K\"ahler} & \cos \alpha = 1\text{”} \\ & & & (1.2) & \\ & & & \text{Wang Proposition 5.2} & \end{array}$$

$$\beta>0$$

4.

4.1 Han--Li--Sun

$$M \quad \text{K\"ahler} \quad \Sigma \quad F: \Sigma \times [0, T) \rightarrow M \quad (1.2) \quad \cos \alpha \geq \delta > 0 \quad \varepsilon_{\text{HLS}} > 0$$

$$\text{Han--Li--Sun Corollary 4.2} \quad X_0 \quad r_{X_0} > 0$$

$$\Psi_p(X_0, T; T-r_{X_0}^2) < 1 + \varepsilon_{\text{HLS}}. \tag{4.1}$$

$$\varepsilon > 0 \quad F \quad [0, T + \varepsilon)$$

$$|A|^2 \leq C/(T-t)$$

$$\text{Han--Li--Sun Corollary 4.2} \quad (4.1) \quad (X_0, T)$$

$$(X_0, T)$$

$$|A| \quad t \uparrow T \quad (x_i, t_i) \quad t_i \uparrow T$$

$$|A|(x_i, t_i) \longrightarrow \infty. \tag{4.2}$$

$$X_i = F(x_i, t_i) \quad M \quad X_i \rightarrow X_0 \in M \quad (X_0, T) \quad r > 0 \quad K < \infty$$

$$|A| \leq K \quad i \quad (X_i, t_i) \tag{4.2} \quad (X_0, T)$$

$$(4.1) \quad \text{Corollary 4.2}$$

$$\sup_{\Sigma \times [0, T)} |A| < \infty. \tag{4.3}$$

$$\text{Han--Li--Sun Theorem 3.1} \quad \varepsilon > 0 \quad [0, T + \varepsilon)$$

$$4.2$$

$$4.1 \tag{1.4}$$

$$\Theta_p^w(X_0, T) < 1 + \varepsilon_{\text{HLS}}, \tag{4.4}$$

$$T$$

$$(4.4) \quad X_0 \quad r_{X_0} > 0 \quad (4.1) \quad 4.1$$

$$\Theta_p^w \leq 1 + \varepsilon_{\text{HLS}}$$

$$5. \tag{2}$$

$$(2.1)\text{--}(2.2) \quad c_j = 1$$

$$\Theta_p^w < 2 \implies \Theta_p^w = 1. \tag{5.1}$$

